Does portfolio optimisation deliver lower risk?

Quantitative Portfolio Optimisation and Risk Analysis
Nine US sector ETFs | Monthly rebalancing | Historical study

Finding: Minimum-variance portfolios had lower held-out volatility than equal weighting, accompanied by more concentrated holdings, more trading and lower compound returns.

Final evaluation: 2020-2025

Portfolio	Annualised volatility	CAGR	Maximum drawdown	Turnover / year
Equal weight	19.82%	12.47%	-36.72%	17.24%
Sample minimum variance	16.82%	5.87%	-32.95%	77.90%
Shrinkage minimum variance	16.84%	7.00%	-32.89%	73.95%

Baseline: 252-day risk window; 5 basis points per dollar bought or sold. All three portfolios use the same 1,508 daily return observations, 2 January 2020 to 31 December 2025. Existing holdings carry into the final period.

The shrinkage portfolio's annualised volatility was **16.84%**, compared with **19.82%** for equal weight: a **15.0% relative reduction**. This is a reduction in volatility, not an increase in investment return.

Its CAGR was 7.00%, versus 12.47% for equal weight. The minimum-variance objective did what it sought to do on the primary risk measure in this sample; it did not maximise growth.

What changed after development?

In 2006-2019, shrinkage volatility was 13.01%, versus 18.10% for equal weight, a 28.1% reduction. Its CAGR was slightly higher. The later evaluation preserved a smaller risk reduction but reversed that return ordering.

Interpretation

Equal weighting is a useful benchmark. Optimisation changes sector exposure as well as estimated risk. A risk-focused portfolio can be less diversified by weight and can miss the returns of sectors receiving small allocations. No universal winner or statistical significance is claimed.

Results are conditional on provider adjusted-price proxies and the stated cost/execution model. Selected independent data checks were completed; remaining source and reinvestment-convention limitations are reported on page 5.

Methods and timing

Fixed rules selected before opening the final evaluation

Universe and comparison

XLB (materials), XLE (energy), XLF (financials), XLI (industrials), XLK (technology), XLP (consumer staples), XLU (utilities), XLV (health care) and XLY (consumer discretionary). The tickers are fixed; holdings and sector definitions evolve.

Equal weight sets each target to 1/9. Both minimum-variance portfolios minimise $\mathbf{w}^T \mathbf{S}\, \mathbf{w}$, subject to weights summing to one and each lying between zero and one. There are no expected-return forecasts, short positions, leverage or extra concentration limits.

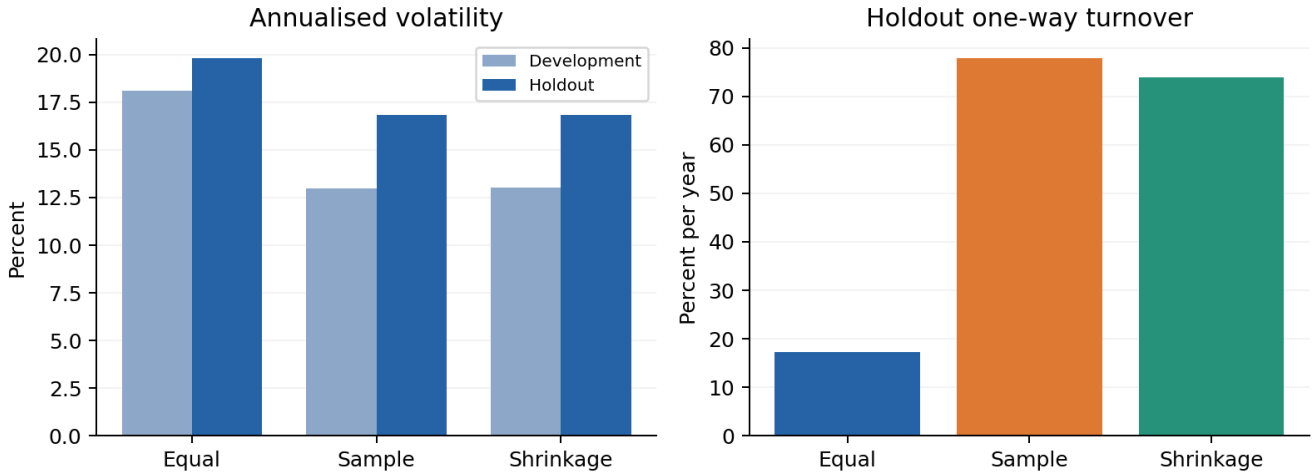
The sample method estimates covariance from past returns. The shrinkage method uses $\mathbf{S_{LW} = (1 - lambda)S + lambda\ mu\ I}$, where mu is average estimated variance. The data determine lambda. The target has equal variances and zero covariances; this is the scaled-identity estimator, not a market-factor model.

Timing example at the final-period boundary

When	What the program does
31 December 2019, after close	Estimate risk using the last 252 available returns and choose targets.
2 January 2020, during the day	The existing portfolio earns the close-to-close return.
2 January 2020, at the close	Trade to the targets and deduct transaction costs.
3 January 2020 onwards	New holdings earn returns and drift until the next rebalance.

This lag separates information availability from execution. Between trades, weights update as w_i,new = w_i,old(1+r_i)/(1+r_portfolio). Equal weighting also rebalances monthly; it is not held at fixed daily weights.

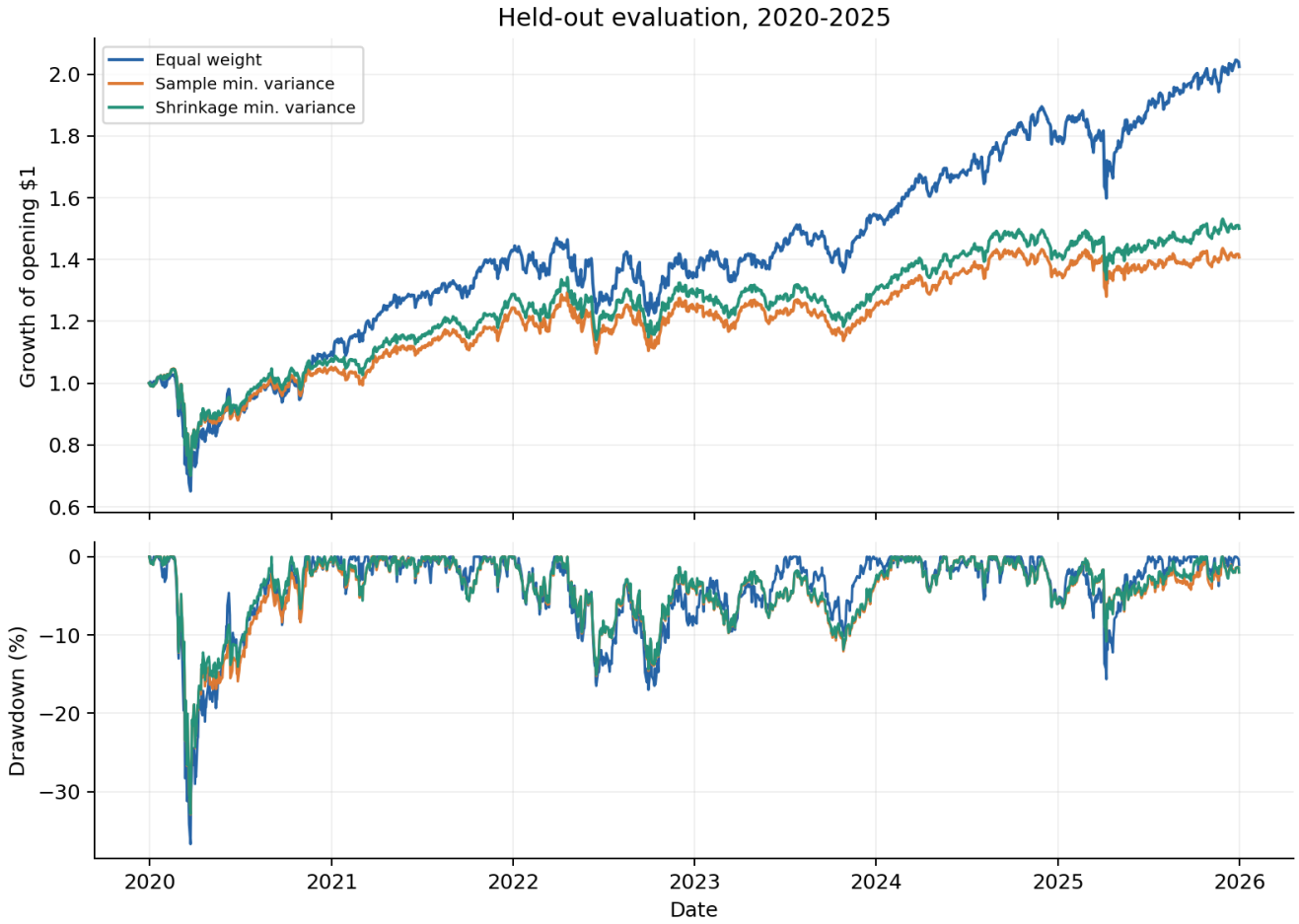
Development versus final evaluation



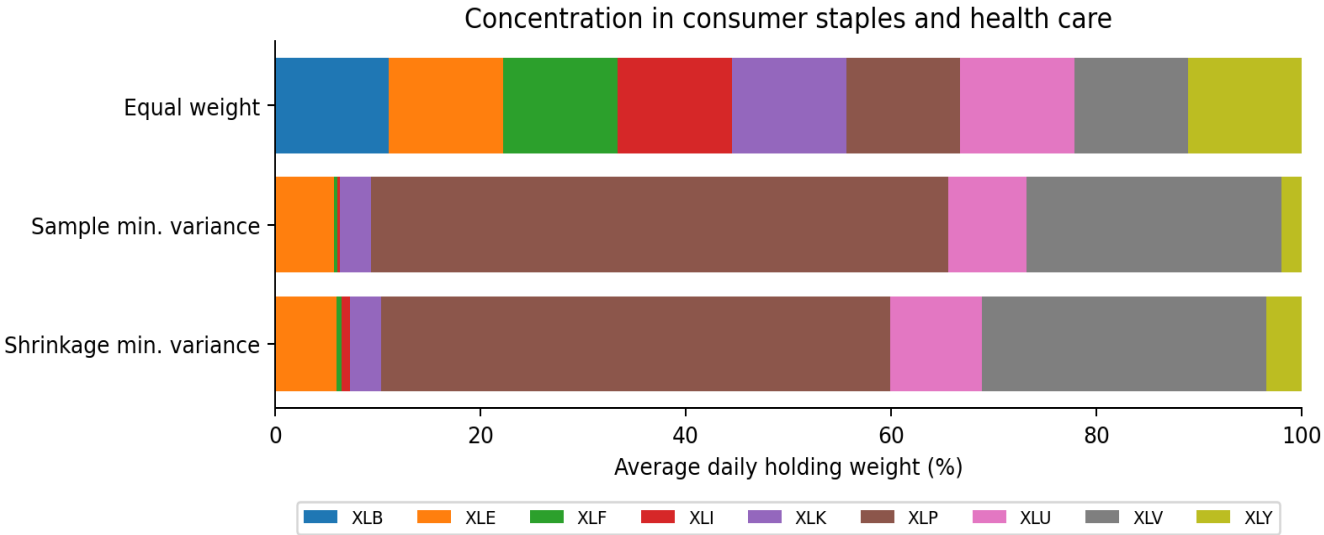
Annualised volatility is sample daily net-return standard deviation times sqrt(252). Trading costs are 0 or 5 bp times the sum of absolute changes from drifted holdings to targets. One-way turnover is half that sum. Establishment costs are included separately from recurring turnover.

The allocation trade-off

Baseline 252-day window, with 5 bp trading costs



Wealth is rebased to one at the end of 2019. Drawdowns use that period-opening value and subsequent peaks. Lower volatility did not prevent substantial losses in the 2020 shock.



Average largest holdings were 57.3% for sample optimisation and 50.5% for shrinkage, compared with 11.6% for monthly equal weighting. Average consumer-staples holdings were approximately 56.2% and 49.6% respectively. These defensive sector tilts accompany the risk result; the study does not isolate a causal benefit from covariance estimation alone.

Robustness and interpretation

One predeclared alternative: 504 past daily returns

Window	Portfolio	Annualised volatility	CAGR	Turnover / year
252	Equal weight	19.82%	12.47%	17.24%
252	Sample min. variance	16.82%	5.87%	77.90%
252	Shrinkage min. variance	16.84%	7.00%	73.95%
504	Equal weight	19.82%	12.47%	17.24%
504	Sample min. variance	17.18%	5.41%	46.84%
504	Shrinkage min. variance	17.20%	5.96%	45.07%

Both windows use identical performance dates. Longer estimation windows reduced turnover and retained a volatility reduction, while producing somewhat higher held-out volatility. This is a sensitivity check, not a search for the best window.

What shrinkage contributed

At 252 days, shrinkage's volatility was almost identical to the sample method (16.84% versus 16.82%). Its turnover was 73.95% versus 77.90%, and its average concentration index was 0.373 versus 0.438. Shrinkage moderated concentration and trading in this sample; it did not improve the primary volatility measure.

What the cost comparison establishes

Zero-cost results are saved alongside 5 bp results. Costs reduce ending wealth on the same holdings path. The assumed rate is illustrative rather than historically measured, and costs use pre-trade NAV with targets applied to the remainder. Spread, market impact, taxes and execution uncertainty are not calibrated.

Limits of the conclusion

The universe is nine long-history US equity sector funds selected with hindsight. It excludes bonds, international assets and some modern sector ETFs. Sector composition changed. Results should not be generalised to all portfolio problems, and do not establish an expected-return advantage.

The final period was withheld from model development within a retrospective exercise. It was not a prospective live test or a claim that the researcher did not know historical events. No naive independent-observation daily t-test or significance claim is used.

A useful interview explanation

"I tested whether estimating minimum-variance weights from past sector returns actually lowered later volatility. It did in the held-out period, but the portfolio became concentrated in defensive sectors and earned less. Shrinkage reduced concentration and turnover, with almost no change in volatility relative to the sample estimate."

Data checks and reproducibility

Selected checks completed; source conventions remain a limitation

The offline snapshot contains 5,535 price dates per ETF from 2 January 2004 to 31 December 2025. Raw-response hashes and all 49,815 adjusted prices reconcile. There is no filling of missing observations. Original prices and strategy settings were unchanged when the final period was opened.

Independent issuer checks

Issuer NAV calendars cover every study date. Three additional issuer valuation dates correspond to confirmed exchange closures. The XLE move on 9 March 2020 is corroborated: -20.141% for the adjusted market-price proxy versus -20.186% for issuer NAV. Five December 2025 2:1 share splits and nine ordinary distribution entries were also checked.

For XLF's September 2016 special distribution, the issuer's 0.139146 entry represents XLRE shares, not dollars. An issuer-NAV event-return reconstruction gives +0.583%, versus +0.638% for the provider's split-style adjusted proxy. Adding the distribution again would double-count it.

An exact independent market-close replication is not claimed. XLF's 2016 provider return is 22.415%; an issuer NAV/ex-date reconstruction gives about 22.659%, while its prospectus reports 22.69%. Price and reinvestment conventions differ. This difference is disclosed rather than erased, and the research findings remain conditional on the provider proxy.

Reproduce and inspect

Run Portfolio_Research.py --period all to reproduce the stored study. The test suite has 17 analytical, accounting, timing and synthetic integration checks. prepare_outputs.py reads existing results to make the presentation charts. verify_data.py reproduces the optional issuer checks without evaluating strategies. Code/data hashes and completion records are included.

Sources

Original provider responses, retrieval URLs and hashes
reference/source_data_audit.json

State Street historical NAV files and distribution data
Issuer data pages and linked XLSX files

XLF in-kind distribution: issuer SEC filing, 2 September 2016
SEC filing

Select Sector SPDR share splits: issuer, 20 November 2025
Issuer announcement

Exchange closure notices
Nasdaq ETA2012-44/45 and NYSE RBO-AMEX-14-04; full links in DATA_REVIEW.md

Ledoit and Wolf (2004), A Well-Conditioned Estimator
Estimator implementation and reference

DeMiguel, Garlappi and Uppal (2009), Optimal Versus Naive Diversification
Review of Financial Studies, 22(5), 1915-1953

AI assisted implementation, checks and explanations. The student must review and understand the methods and findings. This is a bounded empirical study, not a novel optimiser or investment recommendation.